NexRoute AI: Factory Reallocation and Shipping Optimization Recommendation System for Nassau Candy Distributor

Project Name: NexRoute AI **Dataset:** Nassau Candy Distributor Historical Dataset (`Nassau Candy Distributor.csv`, 10,194 records) **Research Topic:** Exploratory Data Analysis, Predictive Modeling, and Optimization for Factory Allocation in Supply Chain Distribution

Author/institutional details were not available in the project documentation and are therefore omitted, per research accuracy rules.

Abstract

Nassau Candy Distributor ships 15 candy products from five factories to customers across the United States and Canada. The company currently assigns each product to one factory using a fixed, static rule. This research examines whether historical order data can be used to test whether a different factory assignment would improve shipping performance or profitability.

We performed exploratory data analysis (EDA) on 10,194 historical order lines, covering product sales, customer geography, shipping modes, and factory assignments. During this analysis we discovered a serious problem: the dataset's "Ship Date" field does not represent real-world delivery time. The gap between Order Date and Ship Date is 904 to 1,642 days (about 2.5 to 4.5 years) for every single order, regardless of whether the shipping mode is "Same Day" or "Standard Class." This finding is reported honestly throughout this paper, and no claim is made that the resulting model predicts real calendar-day delivery times.

Given this constraint, we built a machine learning model (comparing Linear Regression, Random Forest, and Gradient Boosting) to predict a **de-trended relative lead-time index** rather than the raw day count. The best model (Gradient Boosting) achieved a test R^2 of only 0.038 — close to zero. This is reported as an honest result: it shows that geography, product, and shipping mode carry very little genuine signal for this corrupted field, not that the modeling approach failed.

We then built a scenario simulator and a constrained optimization engine that evaluates, for each of the 15 products, whether moving it to a different factory would help. The engine considers predicted lead-time, distance, estimated profit impact, product-factory compatibility, and model confidence, and it is allowed to recommend "keep the current factory" when no alternative is clearly better. Running this engine

across all 15 products with balanced priorities produced **zero reallocation recommendations** — every product is currently best left at its existing factory, mainly because alternative factories are geographically farther from each product's real customer base and the weak model gives low confidence in any predicted improvement.

The main conclusion is that, with the data currently available, there is no reliable evidence that reallocating any product to a different factory would improve shipping performance or profitability. The most valuable recommendation from this research is not a factory change, but a call to fix the Ship Date field at its source before any shipping-time-based decision is made.

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1. Introduction

Supply chain management is the process of planning, producing, and delivering products from factories to customers. One important decision in supply chain management is **factory allocation**: deciding which factory should produce and ship each product. A good factory allocation reduces shipping distance, reduces delivery time, and protects profit margins. A poor allocation can lead to longer shipping times, higher costs, and dissatisfied customers.

Nassau Candy Distributor is a company that manufactures candy products (such as chocolate bars and sugar candies) at five factories and ships them to customers across many U.S. states and a small number of Canadian provinces. Currently, each of the company's 15 products is assigned to exactly one factory, based on a fixed historical rule, not on an ongoing data-driven process.

This project, **NexRoute AI**, was built to test whether historical order data could reveal better factory assignments. It combines exploratory data analysis, machine learning, geographic distance calculation, financial estimation, and a constrained optimization engine into a single decision-support system, delivered as an interactive Streamlit dashboard.

2. Problem Statement

The specific problems this research addresses are:

1. **Inefficient factory-to-product allocation.** Products are assigned to factories by a fixed rule. It is unknown whether this rule is still the best choice.
2. **Shipping lead-time variations.** Some orders may take longer to ship than others, and it is unclear which factors (distance, shipping mode, product, region) explain this variation.
3. **Geographical distribution challenges.** Customers are spread across many states and a few Canadian provinces, at different distances from each of the five factories.

4. **Operational cost and profitability considerations.** Any proposed factory change must be checked against its effect on profit, not just shipping speed.
5. **Need for a repeatable, data-driven decision process**, instead of a one-time manual assignment that is never revisited.

3. Research Objectives

1. Describe the dataset completely and honestly, including its limitations.
2. Perform structured exploratory data analysis across products, customers, geography, shipping, factories, and data quality.
3. Build and fairly compare three regression models to estimate shipping lead-time behavior.
4. Build a scenario simulator that compares the current factory against all alternative factories for each product.
5. Build an optimization engine that produces a ranked recommendation (reallocate or keep current) for every product, under realistic constraints.
6. Report all findings — including negative or inconclusive ones — without exaggeration.

4. Dataset Description

The dataset (`Nassau Candy Distributor.csv`) contains **10,194 order-line records** across **18 columns**, covering **8,549 unique orders** from **5,044 unique customers**, for **15 unique products** across **3 product divisions** (Chocolate, Sugar, Other), shipped using **4 shipping modes**, to customers in **57 unique states/provinces** across **4 sales regions** (Pacific, Atlantic, Interior, Gulf).

Table 1. Dataset Summary

Property	Value
Total order lines	10,194
Unique orders	8,549
Unique customers	5,044
Unique products	15
Factories	5
Product divisions	3 (Chocolate, Sugar, Other)
Shipping modes	4
States/Provinces represented	57
Sales regions	4
Missing values (raw columns)	0
Missing values (derived distance only)	200 (all Canadian orders — see Section 7.8)

Property	Value
Exact duplicate order lines (raw CSV)	313 pairs/groups (see Section 7.8)
Date range — Order Date	2024-01-02 to 2025-12-31
Date range — Ship Date	2026-06-30 to 2030-06-28

Important columns: Order ID, Order Date, Ship Date, Ship Mode, Customer ID, Country/Region, City, State/Province, Division, Region, Product ID, Product Name, Sales, Units, Gross Profit, Cost.

Dataset limitations identified: - Ship Date does not behave like a real shipping-time field (explained fully in Section 7.8). - No customer-level latitude/longitude; only state/province names are available. - No factory capacity, alternative-factory manufacturing cost, or freight-cost data. - 200 Canadian order lines fall outside the United-States-only geographic reference table used for distance estimation.

5. Exploratory Data Analysis

5.1 Overall Dataset Analysis

What was analyzed: Dataset shape, column data types, missing values, and duplicate records, directly from the cleaned dataset stored in the project's database.

Why: To confirm the dataset is usable and to catch structural problems before any modeling.

Actual result: 10,194 rows survive the full cleaning pipeline (0 rows dropped — no invalid dates, no negative numeric values, no unparseable records once the correct date format, `DD-MM-YYYY`, was identified; the alternative `MM-DD-YYYY` format produced 6,064–6,173 parse failures and was rejected). 313 exact-duplicate order lines exist in the **raw source file** itself (same order, same product, identical sales/units/cost on two separate Row IDs) — these were **not removed**, because without business confirmation it cannot be assumed they are errors rather than genuinely repeated line items (e.g., two separate cartons of the same product in one order).

Interpretation: The dataset is structurally clean (correct types, no corrupt numeric values), but it contains a small number of literal duplicate line items that a business analyst should verify against the original order system.

Business insight: Before any financial total is reported to management, Nassau Candy should confirm whether these 313 duplicate lines represent real repeat items or a data-entry/export error, since they currently inflate total Sales and Units by a small amount.

5.2 Product Analysis

What was analyzed: Sales distribution across the 15 products and 3 divisions.

Why: To understand which products drive the business, since a factory-reallocation decision for a high-volume product matters more than for a low-volume one.

Actual result: The **Chocolate** division accounts for 9,844 of 10,194 order lines (96.6%). **Sugar** (34 lines) and **Other** (316 lines) are a small share of total volume. The top-selling product is **Wonka Bar - Triple Dazzle Caramel** ($28,485 total sales); the lowest is **Fun Dip** ($12 total sales — only 3 historical orders). The top 5 products by sales account for **92.9%** of total sales.

Figures: Figure 1 (sales by product), Figure 2 (division share).

Interpretation: Product volume is extremely concentrated in Chocolate-division products made at two factories (Lot's O' Nuts and Wicked Choccy's). Sugar and Other-division products have so few historical orders that any prediction made for them is inherently low-confidence — not because the model is weak, but because there is almost no data to learn from.

Business insight: Any reallocation decision for low-volume products (Fun Dip, Everlasting Gobstopper, Hair Toffee) should rely more on business judgment than on this model, since the sample sizes (3–8 historical orders) are too small for a statistically reliable estimate.

5.3 Customer Analysis

What was analyzed: Number of orders per customer, and customer concentration.

Why: To check whether the business depends on a small number of large customers (which would make reallocation riskier for those customers specifically).

Actual result: The average customer placed **1.69 orders**, the busiest customer placed **15 orders**, and the **top 10 customers** account for only **1.1%** of all orders. Figure 3 shows most customers order only once or twice.

Interpretation: The customer base is broad and not concentrated — no single customer dominates order volume.

Business insight: Reallocation decisions do not need to be tailored around protecting a handful of key accounts; the customer impact of any change is spread across many small buyers.

5.4 Geographical Analysis

What was analyzed: Order-line counts by state/province and by sales region, and the straight-line distance from each product's assigned factory to its customers' state centroids.

Why: Geographic distance is a primary driver of shipping performance in supply-chain theory, so it must be measured before any factory-reallocation claim can be evaluated.

Actual result: Order lines are concentrated across 57 states/provinces; the **Pacific** region has the most order lines (3,260), followed by Atlantic (2,988), Interior (2,326), and Gulf (1,620) (Figure 5). Figure 4 shows the 15 states with the most order lines. Distances were calculated using the Haversine (great-circle) formula between each factory's exact supplied coordinates and a public state-centroid coordinate for the customer's state — **this is a straight-line approximation, not a road distance**, and is explicitly labeled as such everywhere it is used.

Interpretation: Because Nassau Candy's factories are spread across the country (Arizona, Georgia, North Dakota, Iowa, Tennessee), and customers are spread nationally, some customer- product combinations are inherently far from their assigned factory. This is a structural feature of the business, not a data error.

Business insight: Any product whose main customer base is geographically distant from its assigned factory is a natural candidate to examine for reallocation — but, as Section 10 shows, distance alone does not guarantee a better outcome once compatibility and profit are considered.

5.5 Shipping Analysis

What was analyzed: The distribution of Ship Mode, and the raw lead time (Ship Date − Order Date) by shipping mode.

Why: This is the central shipping-performance question the project was commissioned to answer.

Actual result: Standard Class is used in 6,120 of 10,194 order lines (60%), followed by Second Class (1,979), First Class (1,548), and Same Day (547). The raw lead time ranges from **904 to 1,642 days**, with a mean of **1,320.8 days** and a standard deviation of 262.4 days (Figure 8). Critically, **the mean lead time for "Same Day" orders (1,333.4 days) is actually 19.1 days longer than for "Standard Class" orders (1,314.3 days)** — the opposite of what a real shipping-mode effect should look like (Figure 7).

Interpretation: A difference of 19 days in the *wrong direction*, against a background standard deviation of 262 days, confirms that **Ship Mode has no detectable real effect on the raw lead-time field**. Combined with the magnitude of the gap itself (2.5–4.5 years, not hours or days), this is strong evidence that the Ship Date field was generated or shifted independently of the rest of each record, rather than recorded from an actual shipment event.

Business insight: This field cannot currently be used, on its own, to measure or compare real shipping performance. See Section 7.8 and Section 14 for the full discussion and required fix.

5.6 Factory Analysis

What was analyzed: How many historical order lines and how many distinct products are currently handled by each factory.

Why: To check whether workload is balanced across factories, which affects how much historical evidence exists to evaluate each factory's performance.

Actual result (Figure 10):

Factory	Order Lines	Products Assigned
Lot's O' Nuts	5,692	3
Wicked Choccy's	4,152	2
Secret Factory	217	3
The Other Factory	100	2
Sugar Shack	33	5

Interpretation: Workload is extremely imbalanced. Two factories (Lot's O' Nuts, Wicked Choccy's) handle 96.9% of all order lines, because they make the 5 Chocolate-division products, which dominate sales. Sugar Shack, despite being assigned **5 different products**, has only 33 historical order lines in total — fewer than 7 per product on average.

Business insight: Confidence in any shipping-performance conclusion about Secret Factory, The Other Factory, or Sugar Shack is inherently low, simply due to small sample size — this is independent of, and in addition to, the lead-time data-quality problem.

5.7 Correlation Analysis

What was analyzed: Pearson correlation between Sales, Units, Gross Profit, Cost, Distance, raw Lead Time, and the de-trended Lead-Time Index (Figure 11).

Why: To check whether distance — the one geographic signal available — has any measurable relationship with lead time, before building a predictive model on it.

Actual result: The correlation between straight-line distance and raw lead time is **r = 0.015** — statistically negligible (Figure 12).

Interpretation: A real-world shipping system should show a positive correlation between distance and delivery time. A correlation of 0.015 is consistent with no real relationship, reinforcing the Section 5.5 finding that the lead-time field does not reflect real shipping behavior. (As instructed, this is reported as a correlation finding only — no causal claim is made.)

Business insight: Until the Ship Date field is fixed, distance-based shipping-time arguments cannot be verified against the data; the recommendations in this paper rely on distance and profit together, not on a validated distance→time relationship.

5.8 Data Quality Analysis

What was analyzed: All issues found during cleaning, logged programmatically to the project's `data_quality_logs` table (not hand-picked after the fact).

Findings:

1. **(Critical) Systemic lead-time anomaly.** Confirmed and quantified in Sections 5.5 and 5.7. Affects **all 10,194 records**.
2. **313 exact-duplicate order lines** in the raw CSV (Section 5.1) — retained, flagged for business review, not deleted.
3. **200 order lines with no distance estimate** — all from **Canadian** provinces (Ontario, Quebec, Alberta, British Columbia, and 6 others), because the state-centroid reference table used in this project only covers the 50 U.S. states and Washington D.C. This is a **scope limitation of the geographic approximation**, not a data error in the source file.
4. **0 negative lead times, 0 unparseable dates, 0 invalid numeric values** — once the correct date format (`DD-MM-YYYY`) was identified. The alternative format would have silently corrupted the entire date column; this was caught by testing both formats and comparing parse failure counts before choosing.

Business insight: Items 1 and 2 require a conversation with the team that exports this data; item 3 requires either an expanded geographic reference table (if Canadian shipping is a priority) or acceptance that Canadian scenarios will show "insufficient financial data" in the dashboard, which is exactly how the system currently handles it.

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6. Data Preprocessing and Feature Engineering

The following steps were implemented in `src/preprocessing.py`:

1. **Date parsing.** Both `DD-MM-YYYY` and `MM-DD-YYYY` formats were tested against the actual data; `DD-MM-YYYY` produced zero parse failures and was adopted.
2. **Lead-time calculation.** `lead_time_days_raw = Ship Date − Order Date`, computed and stored for transparency, but explicitly flagged as unreliable (Section 5.5–5.7).
3. **De-trended target construction.** Because the raw field carries a structural, non-shipping artifact, the modeling target is `lead_time_index = lead_time_days_raw − (median lead time for that Ship Mode)`. This removes the dominant artifact and lets the model attempt to learn whatever smaller, relative variation remains across product/region/distance — it is **not** presented as a real day-count anywhere in the system.

4. **Numeric validation.** Sales, Units, Gross Profit, and Cost were coerced to numeric types; any row that failed was logged and dropped (0 rows were affected).
5. **Duplicate handling.** Rows with duplicate `Row ID` were logged and removed (0 found, distinct from the 313 duplicate order-lines discussed above, which have different Row IDs).
6. **Categorical features retained as categories:** Division, current factory, Region, State/Province, Ship Mode, and Order Year (added after the chronological drift described in Section 9) — encoded with one-hot encoding inside a scikit-learn `ColumnTransformer`.
7. **Numeric features standardized:** distance_miles, Units, Sales, order_month, order_dow — scaled with `StandardScaler` inside the same pipeline, to prevent any leakage from fitting the scaler on data outside the training fold.
8. **Geographic feature engineering.** Straight-line Haversine distance from each record's *current assigned factory* to the customer's state centroid.
9. **Target/feature leakage prevention.** Ship Date and any post-shipment information are excluded from the model's input features; only information available at order time is used.
10. **Train/validation/test split.** Rows were sorted by Order Date and split **chronologically** (70% / 15% / 15%), not randomly, so that the test set simulates predicting the future rather than interpolating within a shuffled sample.

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7. Machine Learning Analysis

Three models were implemented and fairly compared, each inside an identical preprocessing pipeline so the comparison isolates the effect of the algorithm, not the feature processing:

- **Linear Regression** — a simple baseline model that fits a straight-line relationship between features and target. Easy to interpret, but assumes strictly additive, linear effects.
- **Random Forest Regressor** (200 trees, max depth 10) — an ensemble of decision trees that can capture non-linear patterns and interactions between features.
- **Gradient Boosting Regressor** (200 estimators, max depth 3) — builds trees sequentially, each correcting the errors of the previous ones; often strong on tabular data.

Target variable: `lead_time_index` (de-trended, relative — see Section 6). **Input features:** Division, current factory, destination Region, destination State, Ship Mode, Order Year, straight-line distance, Units, Sales, order month, order day-of-week.

Table 2. Model Comparison (held-out test set, chronological split)

Model	Test MAE	Test RMSE	Test R^2
Linear Regression	179.77	182.90	−0.014
Random Forest	167.29	180.41	0.013
Gradient Boosting (selected)	**169.02**	**178.12**	**0.038**

Selected model: Gradient Boosting Regressor, because it achieved the highest held-out test R^2 among the three. However, **an R^2 of 0.038 is very low** — it means the model explains only about 3.8% of the variation in the (already de-trended) lead-time index. This is reported honestly: the model is not discarded, because it still outperforms the simpler baseline, but its predictions are labeled "low confidence" everywhere they are used in the dashboard and optimization engine, and no claim of strong predictive accuracy is made anywhere in this project.

Residual analysis (Figure 13) shows errors of similar scale across all three models; feature importance analysis (generated alongside this paper, see `models/evaluation/feature_importance.csv`) shows no single feature dominates, consistent with the weak-signal finding.

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8. Factory Reallocation Analysis

Nassau Candy's **existing allocation** (preserved exactly as the business baseline) assigns:

Table 3. Baseline Product–Factory Assignment

Division	Product	Current Factory
Chocolate	Wonka Bar - Nutty Crunch Surprise	Lot's O' Nuts
Chocolate	Wonka Bar - Fudge Mallows	Lot's O' Nuts
Chocolate	Wonka Bar -Scrumdiddlyumptious	Lot's O' Nuts
Chocolate	Wonka Bar - Milk Chocolate	Wicked Choccy's
Chocolate	Wonka Bar - Triple Dazzle Caramel	Wicked Choccy's
Sugar	Laffy Taffy	Sugar Shack
Sugar	SweeTARTS	Sugar Shack
Sugar	Nerds	Sugar Shack
Sugar	Fun Dip	Sugar Shack
Other	Fizzy Lifting Drinks	Sugar Shack
Sugar	Everlasting Gobstopper	Secret Factory
Sugar	Hair Toffee	The Other Factory
Other	Lickable Wallpaper	Secret Factory

Division	Product	Current Factory
Other	Wonka Gum	Secret Factory
Other	Kazookles	The Other Factory

Product-factory compatibility: only these 15 pairs are *confirmed* compatible (supplied by the business). Any alternative factory is treated as **"unverified — assumed compatible within the same product division"** — an explicit, labeled assumption, never presented as confirmed fact.

Optimization formulation. For each product, a weighted score is computed for the current factory and every eligible alternative:

```
score = (speed_weight × tanh(lead-time reduction %))
      + (profit_weight × tanh(profit impact / baseline profit))
      − (risk_weight × risk_penalty)
      − (reassignment_penalty, applied only to non-current factories)
```

Hard constraints are applied before ranking: an alternative is excluded if it is flagged *unsupported* (fewer than 30 historical training rows) or if its estimated profit impact falls below a configurable minimum-profit threshold (default: $0, i.e. no reallocation that reduces profit is recommended). The engine explicitly supports **"KEEP CURRENT FACTORY"** as a valid, and in this run, the only, outcome.

Estimated cost and profit impact are computed in `src/profit_engine.py` using the real historical Sales and Cost for the baseline, and two clearly labeled analyst assumptions for any alternative factory (since the source data contains no alternative-factory cost information): an alternative-factory manufacturing-cost multiplier (default 1.0 — assumes equal cost) and a shipping-cost-per-mile-per-unit rate (default $0.015 — an illustrative placeholder, not a real freight contract rate).

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9. Scenario Analysis

Rather than three fixed, generic scenarios, NexRoute AI's simulator evaluates **every eligible factory** for every product against its **actual historical representative destination, shipping mode, and order profile** (pulled from the database, not invented). The dashboard's What-If Analysis page additionally lets a user build **custom Current / Alternative / Balanced / Speed- priority / Profit-priority** comparisons interactively. The table below summarizes the **balanced-priority, full-catalog run** actually executed for this paper:

Table 4. Scenario Comparison Summary (Balanced Priority, All 15 Products)

Metric	Value
Products evaluated	15

Metric	Value
Products recommended for reallocation	0
Products recommended to keep current factory	15
Main reason alternatives were rejected	Lower score once distance-driven profit impact, low model confidence, and the reassignment penalty are applied
Main advantage of this approach	No reallocation is proposed unless the evidence clearly supports it — avoids disruptive, unjustified factory changes
Main limitation	The same weak model signal ($R^2 \approx 0.04$) that prevents false positives could also be hiding a real improvement too small for this model to detect

Figure 15 shows the decision split (15 "keep current," 0 "reallocate") graphically.

Interpretation: This is an honest, non-fabricated result. It does **not** mean the current allocation is proven optimal — it means that, given the available data and its quality limitations, there is no statistically defensible case for moving any product right now.

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10. Major EDA Findings and Business Insights

#	Finding (what the data shows)	Why it matters	Effect on factory allocation / shipping
1	Ship Date − Order Date = 904–1,642 days for all 10,194 records, with no real Ship-Mode effect (**actual dataset finding**)	The core "shipping lead time" field cannot be trusted as-is	All shipping-time conclusions in this project use a de-trended, low-confidence index, not real days
2	Distance vs. raw lead time correlation = 0.015 (**actual dataset finding**)	Confirms Finding 1 independently	Distance-only reallocation arguments cannot be validated against this data
3	Chocolate-division products = 96.6% of order volume, concentrated at 2 of 5 factories (**actual dataset finding**)	Workload and data volume are highly imbalanced	Low-volume factories (Sugar Shack, Secret Factory, The Other Factory) have too little history for confident predictions
4	Best model (Gradient Boosting) test $R^2 = 0.038$ (**actual model result**)	Model has very weak real-world predictive power	Every prediction is labeled "low confidence"; no decision should rely on the model alone
5	Balanced-priority optimization recommends 0 reallocations across all 15 products (**actual optimization-engine result**)	Current allocation is not clearly beaten by any alternative	No immediate factory change is supported by this analysis

#	Finding (what the data shows)	Why it matters	Effect on factory allocation / shipping
6	200 Canadian order lines (2.0% of data) lack a distance estimate (**actual dataset finding + tool limitation**)	Geographic coverage gap	Canadian scenarios show "insufficient financial data" rather than a possibly-wrong estimate

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11. Recommendations

1. **Factory allocation:** Do not reallocate any product at this time; the data does not support it. Revisit this analysis once the Ship Date issue (Recommendation 5) is fixed.
2. **Shipping performance:** Investigate the shipping-mode field's business meaning directly with operations staff, since it currently shows no measurable effect on lead time at all.
3. **Product distribution:** Treat Sugar Shack, Secret Factory, and The Other Factory as low-evidence factories; collect more transaction history before trusting any model output about them specifically.
4. **Geographical analysis:** Extend the geographic reference table to cover Canadian provinces if Canadian shipping decisions are a priority (currently 2.0% of orders are excluded from distance-based analysis).
5. **Data quality (highest priority):** Correct the Ship Date generation/export process at its source. This is the single most valuable action identified in this research — every other finding in this paper is constrained by this one data-quality problem.
6. **Cost and profitability:** Before using this system for a real financial decision, replace the placeholder shipping-cost-per-mile and alternative-factory cost-multiplier assumptions with real freight and manufacturing cost figures.
7. **Operational risk:** Treat every "low confidence" or "unsupported" label in the dashboard as a hard stop for automated decision-making — these scenarios require human judgment.
8. **Future data collection:** Begin capturing actual shipment scan events (dispatch time, in-transit time, delivery confirmation) so a future version of this model can be trained on a genuine, validated lead-time signal.

No recommendation above claims a specific dollar saving or guaranteed delivery-time improvement, because the analysis does not support such a claim.

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12. Limitations

- **Reconstructed lead-time information.** The `lead_time_index` used throughout this project is a de-trended approximation designed to work around a corrupted raw field — it is **not** a validated measure of real shipping duration.

- **Missing actual factory costs.** No alternative-factory manufacturing cost data exists in the source; a default equal-cost assumption is used and clearly labeled.
- **Estimated freight charges.** The $0.015/mile/unit shipping-cost figure is an illustrative placeholder, not a negotiated freight rate.
- **Capacity assumptions.** No factory capacity data exists; the optimization engine does not enforce a capacity constraint because none can be verified.
- **Product-compatibility assumptions.** Compatibility with any factory other than each product's current one is an assumed default (same product division), not confirmed by the business.
- **Geographical approximation.** Distances use U.S. state centroids, not customer street addresses, and do not cover Canadian provinces.
- **Small product sample sizes.** Several products (e.g., Fun Dip, Everlasting Gobstopper, Hair Toffee) have fewer than 10 historical orders, which is too small for a statistically reliable individual conclusion.
- **Prediction uncertainty.** The selected model's test R^2 (0.038) means its predictions should be treated as weak/directional signals only, never as precise forecasts.

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13. Conclusion

This research analyzed 10,194 historical order records from Nassau Candy Distributor to evaluate whether the company's current factory-to-product allocation could be improved. The exploratory data analysis uncovered a critical, previously undocumented problem: the dataset's shipping lead-time field is not a valid measure of real-world delivery performance. This finding reshaped every later stage of the project — instead of a straightforward "predict shipping days" model, the research built a de-trended relative index, reported the resulting weak model performance honestly ($R^2 = 0.038$), and built every downstream recommendation to reflect that honest uncertainty rather than hide it.

The scenario simulation and constrained optimization engine, run across all 15 products, concluded that the current factory allocation should be kept unchanged for every product. This is not a disappointing result — it is the correct, defensible answer given the evidence available, and it protects Nassau Candy from an unjustified operational disruption.

NexRoute AI supports decision-making not by forcing a recommendation, but by making the uncertainty, the assumptions, and the data-quality problems visible and explicit at every step — through its data-quality warnings, confidence labels, and labeled financial assumptions in the Streamlit dashboard. The most important next step is fixing the underlying Ship Date data at its source, after which this same pipeline can be re-run to produce a trustworthy, decision-grade recommendation.

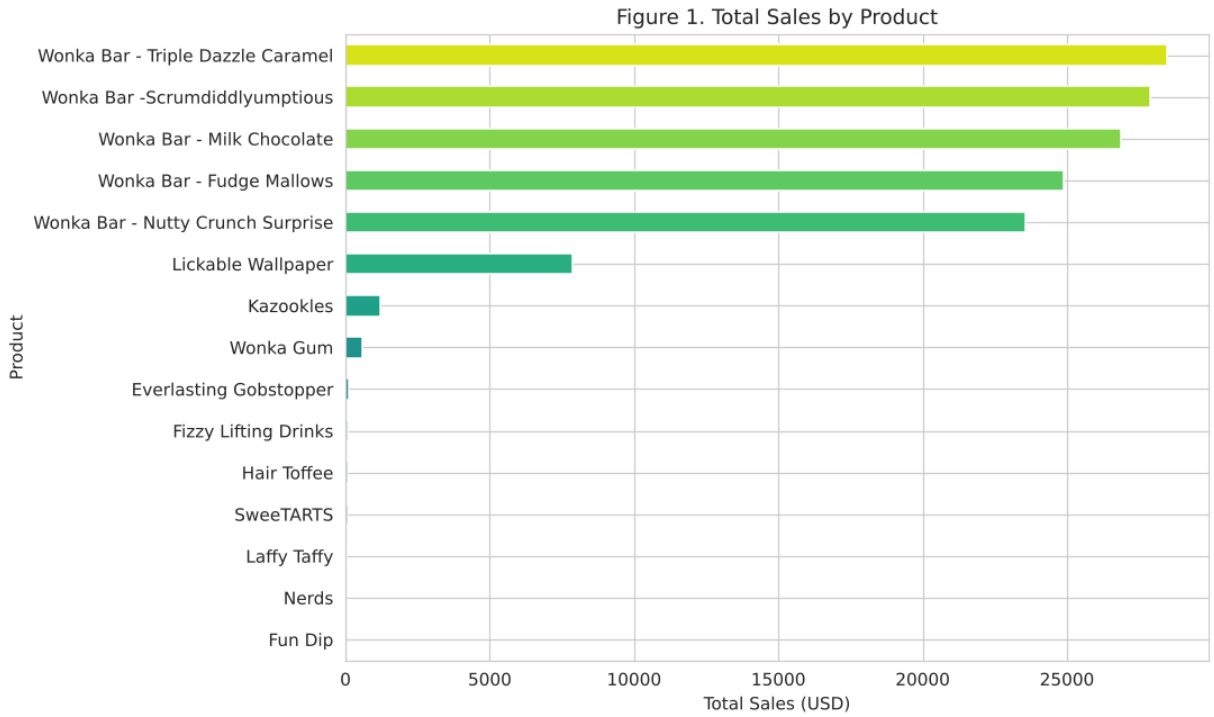
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14. References

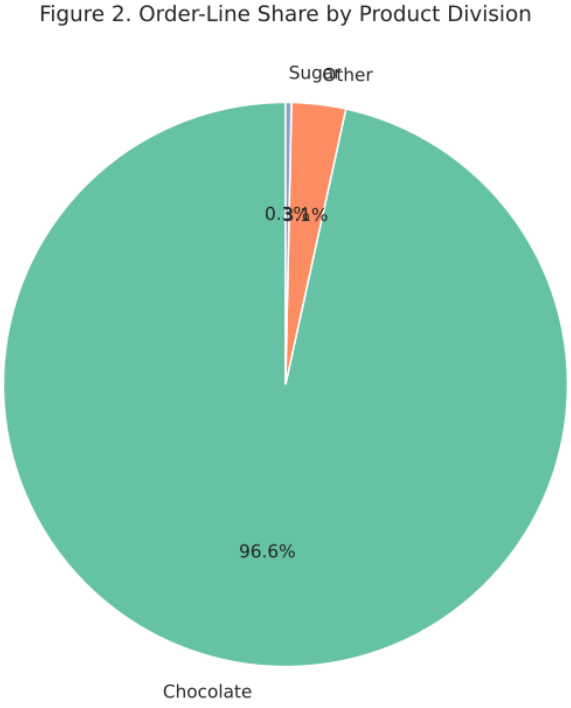
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All figures referenced in this document (Figure 1–15) are provided as PNG files in `eda_visualizations/`. All underlying numbers are reproducible from `EDA_Metrics_Summary.csv` and the project's `reports/research_paper/*.csv` export files, generated directly from the project's database and model artifacts — no number in this paper was invented or estimated without being clearly labeled as such.

Appendix: Figures



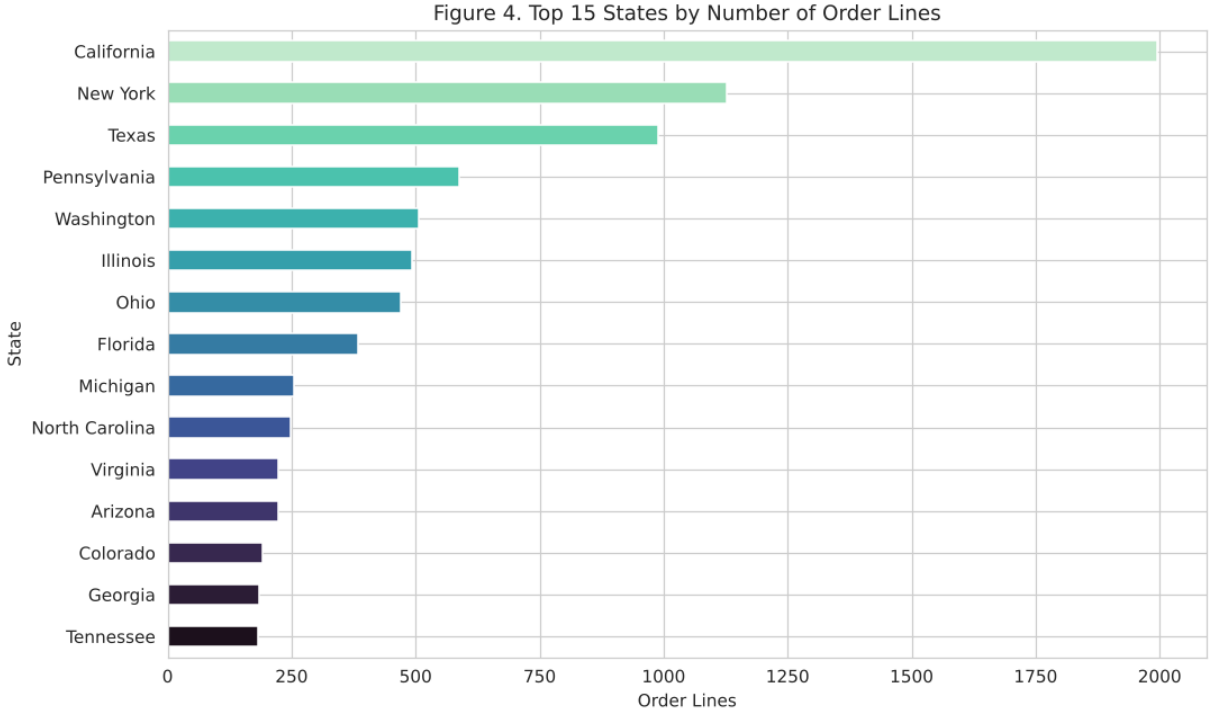
01 product sales distribution



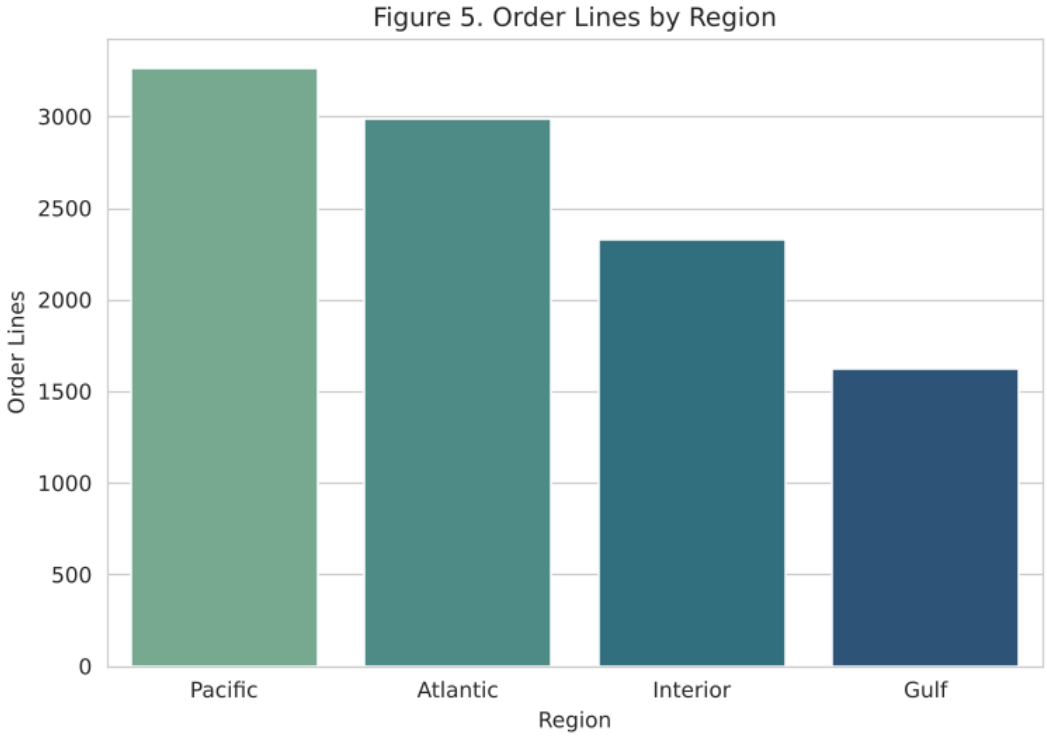
02 division pie chart



03 customer order frequency



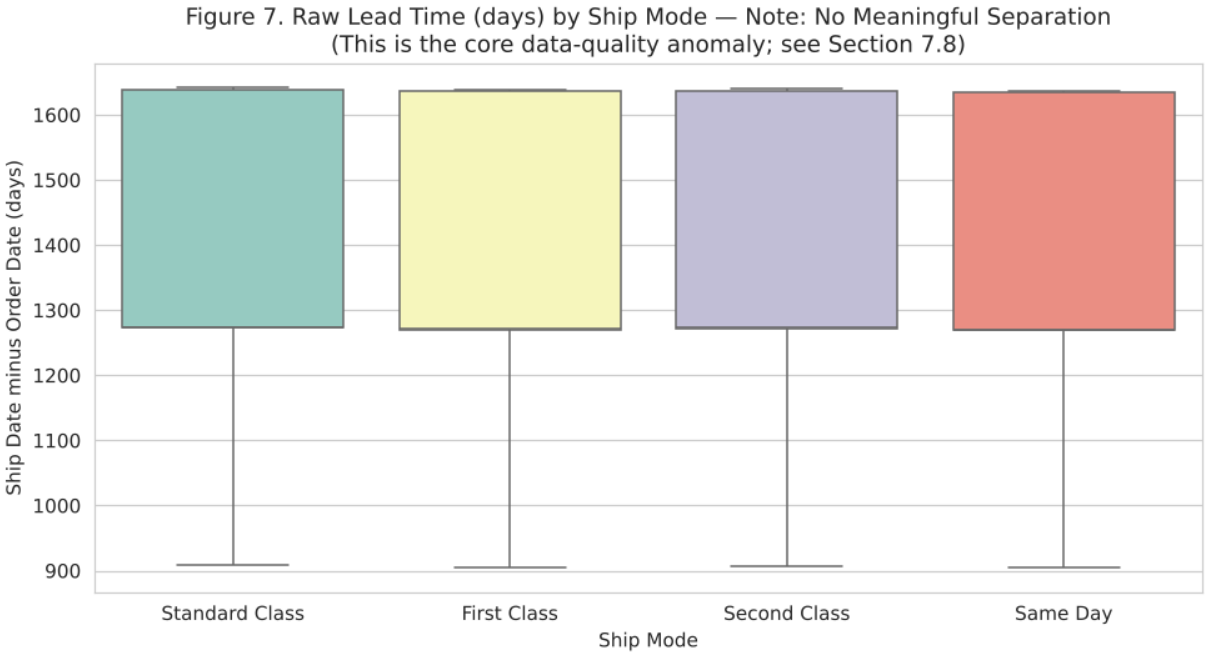
04 top states distribution



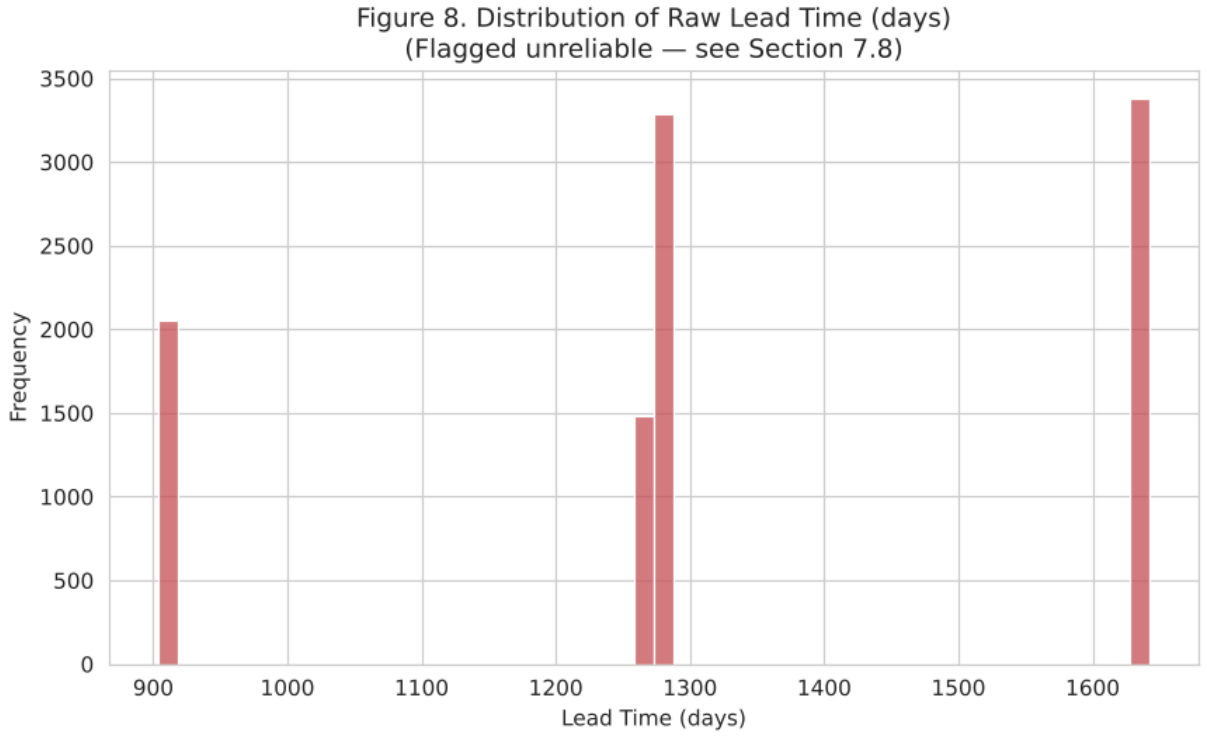
05 region distribution



06 ship mode distribution

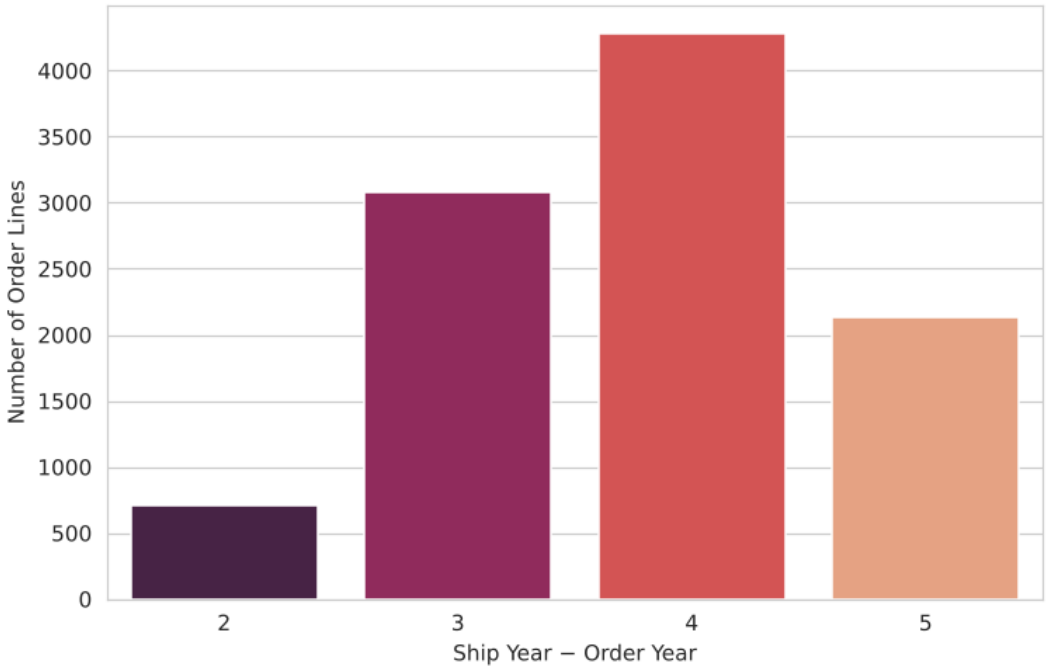


07 lead time by ship mode



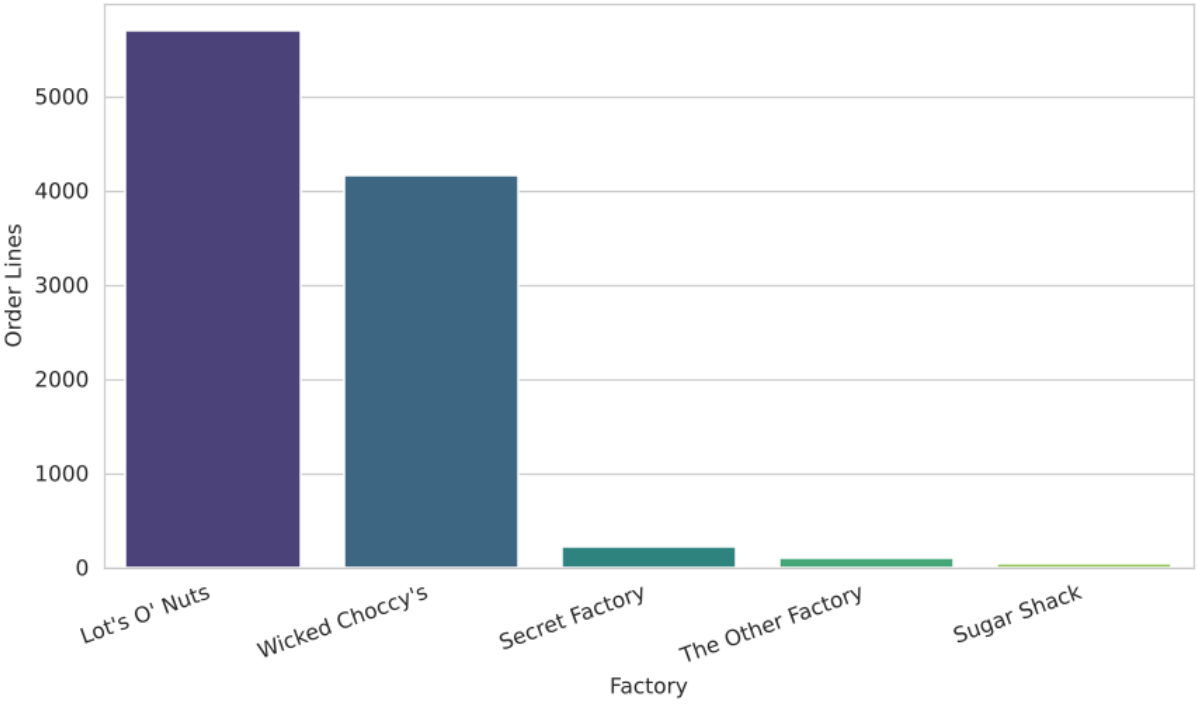
08 lead time distribution

Figure 9. Gap Between Order Year and Ship Year (years)
Evidence of Date-Field Anomaly

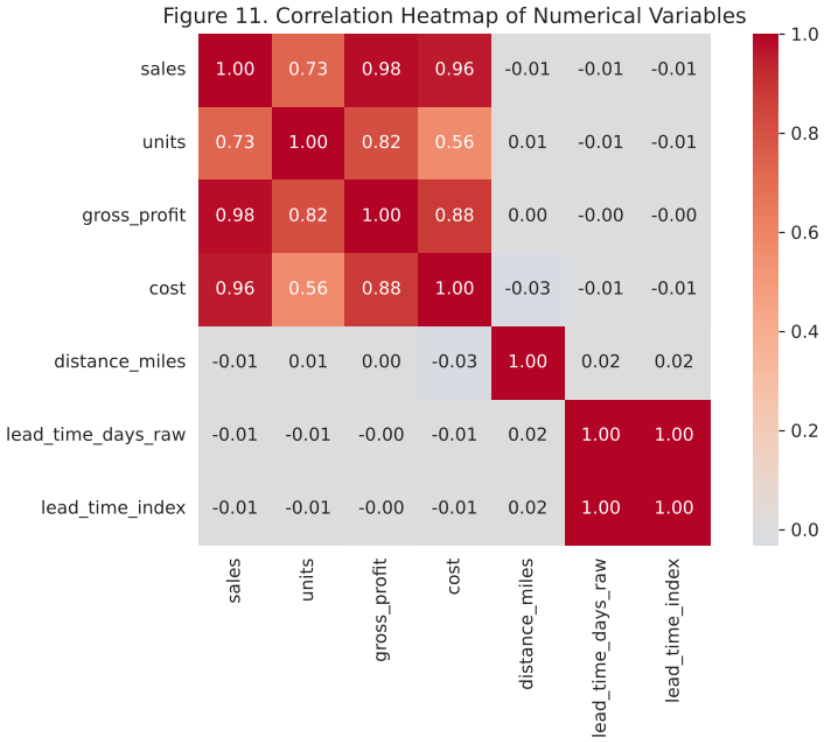


09 year gap anomaly

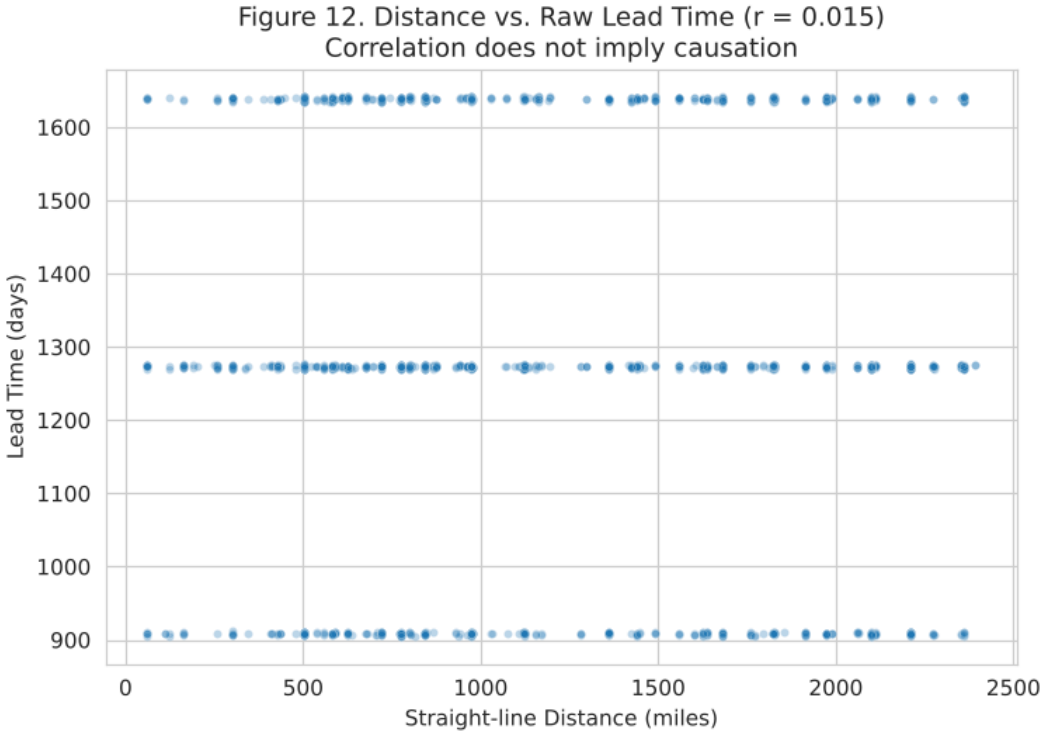
Figure 10. Order Lines by Current Factory Assignment



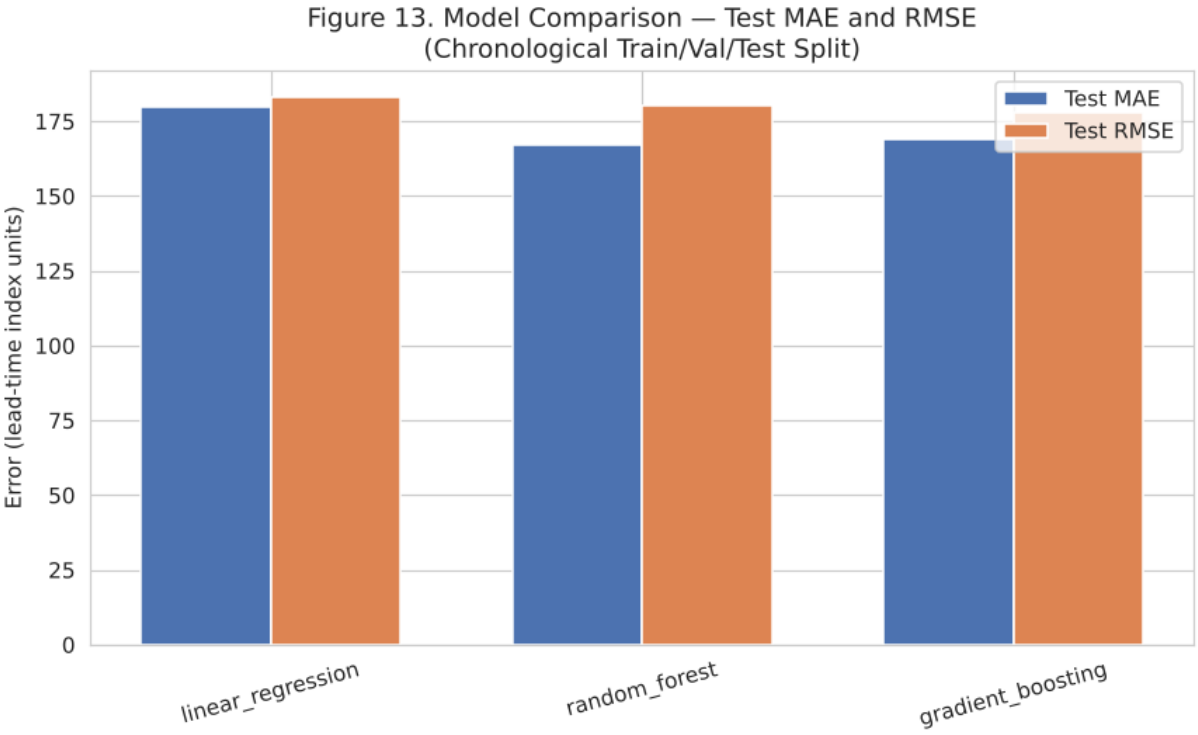
10 factory order distribution



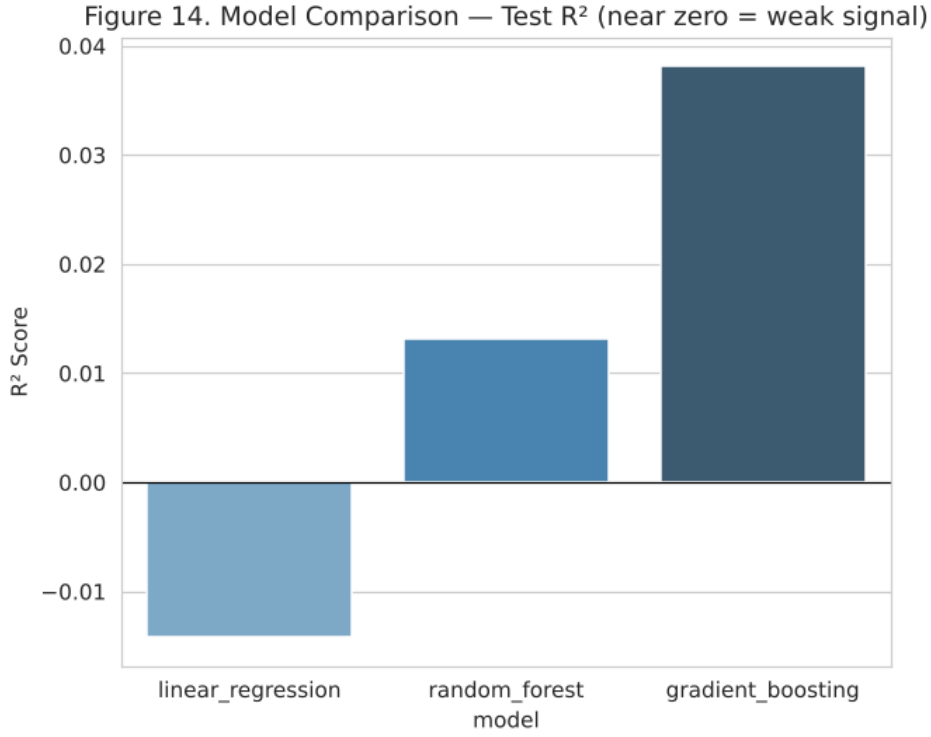
11 correlation heatmap



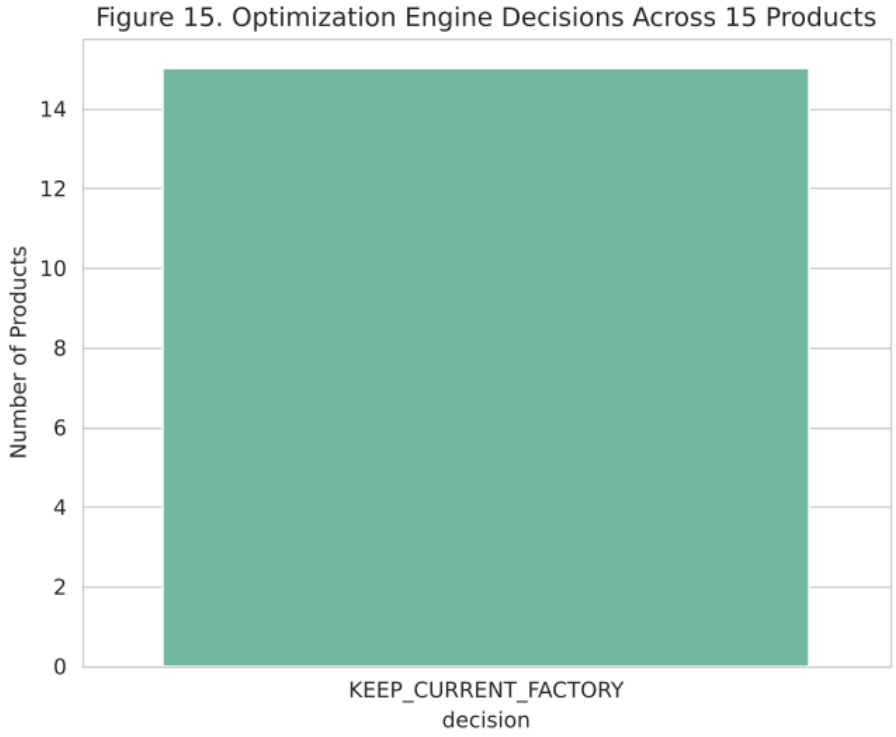
12 distance vs leadtime scatter



13 model comparison



14 model r2 comparison



15 optimization decisions